MECH MAINT- TURBINE Department
SIMHADRI

Ref:SMTPP/MECH MAINT-/2022-23/MM/TMD/109272

Date: 17/05/2022(18:22:55)

Subject: Revision of LMI "Hydro Test of Lube & jacking Oil Lines of KWU Turbines" LMI NO. LMI/OGN/OPS/MECH/043- Reg

The LMI: "Hydro Test of Lube & jacking Oil Lines of KWU Turbines" with LMI NO. LMI/OGN/OPS/MECH/043 is revised by CC-OS.

Following changes were done in LMI as per revised OGN:

- Drg-3 added.
- Point 5, s.no: iv revised, s.no: xv added.
- Point 7, s.no: vi revised.

Put up for Approval of Competent authority.

Encl:

- Earlier approved LMI.
- New LMI
- Revised OGN

Submitted by:

रामबाबू वानापल्ली - उप महाप्रबंधक

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On:17/05/2022(18:22:55) **Ref:**SMTPP/MECH MAINT-/2022-23/MM/TMD/109272

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LMI was studied and forwarded for further approval

Submitted by:

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On:24/05/2022(19:33:21) **Ref:**SMTPP/MECH MAINT-/2022-23/MM/TMD/109272

LMI reviewed and found OK. Revised LMI may please be accorded approval.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoor - 005637 (GENERAL MANAGER ,MAINTENANCE)
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On:25/05/2022(09:58:59) **Ref:**SMTPP/MECH MAINT-/2022-23/MM/TMD/109272

कृपया अनुमोदन प्रदान किया जाए।

May be approved please.

Submitted by:

राजेश कुमार झा - महाप्रबंधक

Rajesh Kumar Jha - 003323 (GENERAL MANAGER ,O & M)
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On:25/05/2022(11:43:44) **Ref:**SMTPP/MECH MAINT-/2022-23/MM/TMD/109272

Approved & Submitted by:

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On:25/05/2022(16:52:36) **Ref:**SMTPP/MECH MAINT-/2022-23/MM/TMD/109272

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/043/

ISSUE NO.1; REVISION NO. 04; DATE: MAY'2022

Hydro Test of Lube & Jacking Oil Lines of KWU Turbines

Approved for

Implementation____________________________

HOP (SIMHADRI)

Date: ______________

SIMHADRI

LMI APPROVAL DOCUMENT

TITLE OF LMI	HYDRO TEST OF LUBE & JACKING OIL LINES OF KWU TURBINES
a) DOCUMENT CODE b) REVISION / ISSUE NO & DATE	LMI/OGN/OPS/MECH/043/ ISSUE NO.1, REV.04, MAY'2022
ORIGINAL / REF. DOCUMENT	OGN/OPS/MECH/043 REV.01, APR'2022
PREPARED BY	DGM (MM-TMD)
CHECKED BY	AGM (MM-TMD)
RECHECKED BY	GM (Maint)
REVIEWED BY	GM (O&M)
APPROVED FOR ISSUE	HOP
INDEX LIST	ENCLOSED
DISTRIBUTION LIST	ENCLOSED

INDEX

1.0 INTRODUCTION

In order to improve the reliability of TG lube oil and jacking oil lines of all four units of NTPC, Simhadri, which have run for more than 150000 hrs, it is proposed to do a hydro test of oil lines in lube oil and jacking oil systems. When the machines are on load, the leakages from joints and flanges of lube oil and jacking oil systems, may lead to forced shutdowns and create a potential fire hazard. This test is designed to be carried out during capital overhaul of turbine to identify the defects and repair them during overhaul.

2.0 SUPERSEDED DOCUMENTS

REV: 03, DATE: FEB'2022.

3.0 REFERENCE DOCUMENT:

OGN/OPS/MECH/043, Rev.01, Date: APR'2022

4.0 HYDRO TEST PRESSURE

The hydraulic test of TG Lube oil lines and jacking oil lines is carried out at pressures mentioned as below.

SNO	SYSTEM	STANDARD PRESSURE	TEST PRESSURE	DURATION
1	Lube oil system with coolers	2.5ksc	12ksc	10 min
2	Jacking oil system	120ksc	180ksc	10 min

5.0 PREPARATION ACTIVITY FOR HYDRO TEST OF LUB OIL LINES

Following preparation has to be carried out during capital overhaul of TG set after stopping of lub oil system before carrying out hydro test of lube oil lines. (Refer DRG-1).

i. All the lube oil inlet valves/ throttles to the TG bearings shall be kept closed at the bearing inlet. Before closure of throttles, heights of individual throttles are to be measured and recorded.

ii. MOP suction line is dummied and a vent shall be provided for venting.

iii. The lube oil header dummy at the exciter end is connected with a temporary loop to drain.

iv. Inter connection line (1/2" dia) from jacking oil header to lube oil discharge header is to be erected with two valves in the MOT room or near exciter end bearing

v. Oil inlet and return from both injectors to lube oil system are to be blanked.

vi. Both the coolers are to be lined up for hydro test.

vii. Lube oil/ thrust bearing filter internals are to be removed.

viii. AOP 1&2 discharge valves are kept closed. (In many cases where NRVs are provided which isolate the pump)

ix. All the pressure and instrument tapping are to be isolated at the root.

x. Calibrated gages are to be provided on the header to show pressure, one near the pumping source and other at the farthest location at lub oil supply line near TG bearing no.7.

xi. Venting arrangements are to be provided at the blind ends with valve.

xii. TG turning gear valve is kept closed.

xiii. Healthiness of all hangers and supports in lube oil lines are to be ensured before test.

Resp: HOD (TMD)

xiv. List of the dummy and bypass provisions to the normal system is made to be recorded jointly with operation.

Resp: HOD (TMD) / HOD (Operation)

xv. Arrangements should be made to meet any exigency created due to oil leakage. Sufficient no. of empty oil drums should be kept in MOT room and oil canal. Fire personnel also to be deployed during testing. Proper illumination should be ensured in oil canal & MOT Room.

Resp: HOD (TMD) / HOD (Operation)

6.0 PREPARATION ACTIVITY FOR HYDRO TEST OF JACKING OIL LINES

Following preparation has to be carried out during capital overhaul of TG set after stopping of lub oil system before carrying out hydro test of jacking oil lines. (Refer DRG-2).

i. All the jacking oil inlets/throttles to the TG bearings are kept closed at the bearing inlet.

ii. Jacking oil bearing filter internals to be removed

iii. The jacking oil header dummy at the exciter end is connected with a temporary loop to drain.

iv. All the pressure and instrument tapping are to be isolated at the root

v. A calibrated gage is to be provided on the header to show pressure

vi. Venting arrangements are to be provided at the blind ends with valve

vii. Healthiness of all hangers and supports in lube oil lines are to be ensured before test.

Resp: HOD (TMD)

viii. List of the dummy and bypass provisions to the normal system is made to be recorded jointly with operation

Resp: HOD (TMD) / HOD (Operation)

7.0 METHOD FOR LUBE OIL HYDRO TEST

i. DCEOP is run and the system is filled.

ii. Venting of the total system is carried out

iii. DCEOP is stopped

iv. One JOP is started with pressure regulating valve fully opened.

v. Pressure is raised to 12ksc by opening interconnection and pressure is held for 10 min.

vi. System is checked for leakages. If leakages are observed depressurize the system and attend leakages and repeat the test

Resp: HOD (TMD) / HOD (Operation)

8.0 METHOD FOR JACKING OIL HYDRO TEST:

i. DCJOP is run and the system is filled by keeping PRV in release condition.
ii. Venting of the total system is carried out and the drain interconnection is closed. Refer 6 (iii)
iii. Raise the pressure to 180 ksc with the help of PRV. Pressure is held for 10 min and system is checked for leakages.
iv. If leakages are observed depressurize the system and attend leakages and repeat the test.

Resp: HOD (TMD) / HOD (Operation)

9.0 COMPLETION CRITERIA

The hydro test is completed when there are no oil leakages in jacking/lube oil systems**.**

Resp: HOD (TMD)/HOD (Operation)

10.0 NORMALISATION

All points mentioned in the dummy and bypass list to be normalized. Joint protocols to be made with operation.

Resp: HOD (TMD)/HOD (Operation)

11.0 EXCEPTION REPORTING

Any exception to the instructions specified in the LMI are to be reported as per the following format to HOD (O&M)

Resp: HOD (O) / HOD (TMD)

Description of Exception	Unit No.	Lube oil / Jacking oil system	Remarks

12.0 RECORD KEEPING

The records to be maintained under this LMI include

a) Joint protocol in between Operation and TMD consisting of Date of test, defects, if any.
b) Exception reporting.

The above records are to be maintained by Turbine maintenance / Operation.

13.0 REVIEW

Head of the project (HOP) shall review this LMI note once in two years or as and when the reference document is revised.

REVISION INDEX

(FOR CHANGE / REVISION)

SL NO	REV NO	DATE	DETAILS	APPROVED BY
01	**01**	**OCT'2018**	**Point No:1** Criteria for doing hydro test added. i.e. Units which have run for more than 150000 hrs added.	
02	**02**	**JAN'2020**	**Point no-13:** Review period changed to two yearly basis from three-yearly basis. **Point no-2**: Revision no and date changed	
03	**03**	**JAN'2022**	**Drg-1 & 2** added. HOD (TMD) added at S.NO: 7 & 8	
04	**04**	**MAY'2022**	**Drg-3** added. Point 5, s.no: iv revised, s.no: xv added. Point 7, s.no: vi revised.	

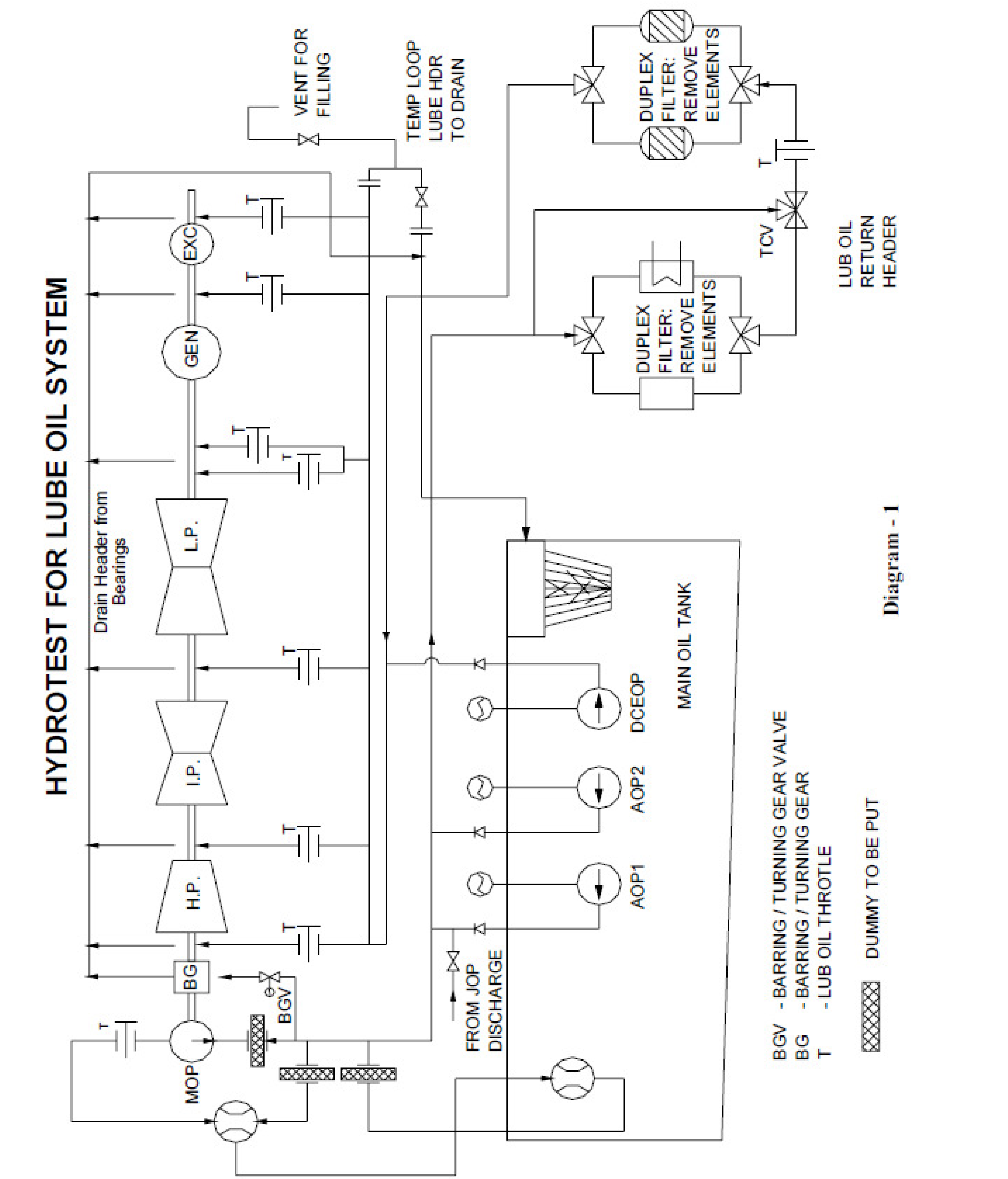
DISTRIBUTION LIST

SL NO	DATE OF ISSUE	ISSUED TO	COPY NO
		HOD (O&M)	
		HOD (Maint)	
		HOD (O)	
		HOD (TMD)	

RECORD OF HYDRO TEST

RESP: HOD(TMD) / HOD (O)

SL. NO	UNIT	LUBE OIL /JACKING OIL SYSTEM	TEST PRESSURE	DURATION



HYDROTEST FOR LUBE OIL SYSTEM
Drain Header from Bearings
VENT FOR FILLING
TEMP LOOP LUBE HDR TO DRAIN
DUPLEX FILTER: REMOVE ELEMENTS
DUPLEX FILTER: REMOVE ELEMENTS
TCV
LUB OIL RETURN HEADER
EXC
GEN
L.P.
I.P.
H.P.
BG
BGV
MOP
MAIN OIL TANK
DCEOP
AOP2
AOP1
FROM JOP DISCHARGE
BGV - BARRING / TURNING GEAR VALVE
BG - BARRING / TURNING GEAR
T - LUB OIL THROTLE
DUMMY TO BE PUT
Diagram - 1

NTPC	DRAWING-2	SIMHADRI
TITLE: HYDRO TEST OF LUBE & JACKING OIL LINES OF KWU TURBINES.		
REF NO. LMI/OGN/OPS/MECH/043; REV:04; DATE: MAY'2022		**PAGE: 11 OF 12**

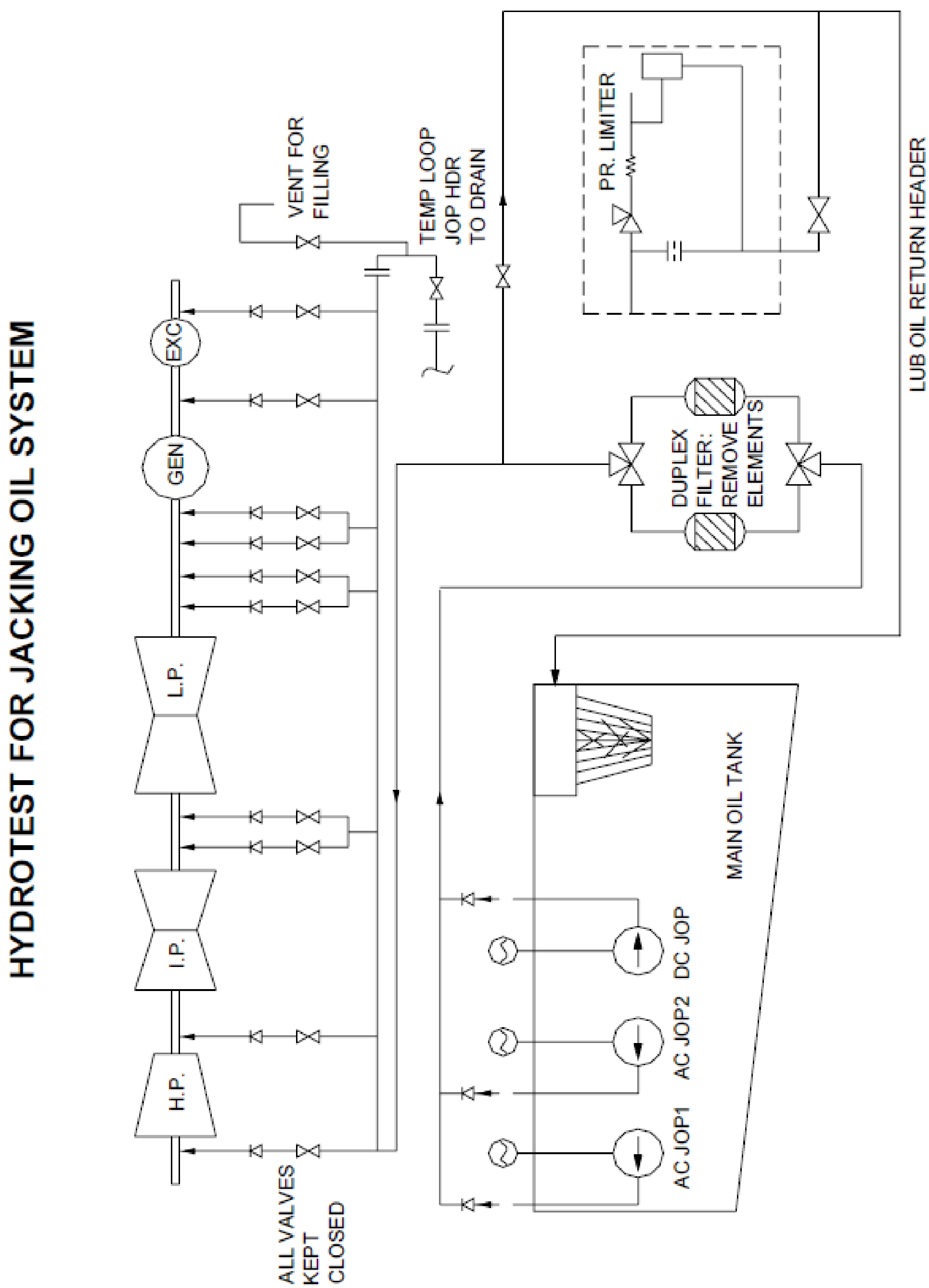


Diagram - 2

DRAWING NO.3

Hydro test device with vent should be placed after removal of MOP, Oil inlet line should be dummied. Rings may be required for leveling purpose.

Interconnection of JOP discharge to Lub oil system for Pressurinsing

Both injectors (MOT & Oil canal) motive lines should be dummied